

How evenly can the burden be shared? Our package raises EUR 103 bn (.71% of EU GNI), but its cost ranges from .34% of GNI in Slovakia to 1.63% in Cyprus (column (1) of Table 2). This dispersion pushes countries that collect more revenue than the EU average to call for a harmonised, domestic implementation of these new taxes, with each Member State retaining the revenues it collects, rather than establishing them as EU own resources. How much of this dispersion can be avoided by changing the relative weights of these six taxes? Holding the total revenue constant, we let each tax rate vary between zero and four times the rate proposed in the baseline scenario and look for the combination that spreads the burden most evenly, assuming that each tax's revenue is proportional to its rate. We use two criteria: minimising the highest national cost (*Minimax*), and minimising the sum, over the 27 Member States, of squared deviations of cost from the EU average (*Sq. dev.*). We also compute two variants of the latter criterion: one requires the five frugal countries (Austria, Germany, Ireland, the Netherlands and Sweden) to bear a cost below the EU average (*Frugal*); the other maximises the smallest State budget gain (*Max. gain*), that accounts for foregone revenue of existing domestic taxes (column (5) of Table 2). Table 3 reports the results, first with a single luxury VAT rate, then with the luxury tax split into seven product taxes with their own rates. in proportion to its category-specific base (see the luxury appendix).

Recomposing the package cuts the dispersion of costs roughly in half: the highest national cost falls from 1.63% to .88% of GNI, or the root mean squared deviation (RMSD) from the EU average from .31 to .15 points. Both criteria pull the same lever: the luxury VAT and the aviation tax concentrate their cost on small and tourism-intensive economies, so both cut them to zero or nearly so and shift the burden onto the advertising tax, the digital services tax and the wealth tax, whose revenues are the closest to proportional to GNI. Requiring the frugal countries to pay less than the average costs about half of this gain: it rules out the advertising tax, whose base is concentrated in Germany, Sweden and the Netherlands, and raises the aviation tax to 36%, leaving Cyprus at a cost of 1.27% of GNI. Maximising the smallest State budget gain lifts France, the least favoured Member State, from a State gain of .24% to .45% of GNI, but it requires giving up the wealth and luxury taxes, which finance the new funds, and it makes costs less even again. Taxing luxury categories at separate rates recovers much of the loss in the case favoring frugal countries: taxing cars, gourmet food and design while exempting hospitality and fine wines, whose bases are the most unevenly spread, brings the root mean squared deviation down to .14 points in the frugal variant, against .22 when the luxury tax carries a single rate.

Among these variants, the minimax criterion with the luxury tax split by category (in bold in Table 3) may be the best compromise. It caps the cost of the package at .85% of GNI, the lowest ceiling of all the combinations considered, the DST stays at a reasonable rate, and the French budget gain is lifted from .24 to .31% of GNI (compared to .41% for countries without foregone revenue). Finally, setting an instrument aside in the EU package does not mean giving up on it: the higher rates that this exercise discards, in particular on aviation and luxury, can still be implemented through an EU-wide harmonisation policy, with the revenue retained by the Member States that collect it.

Table 3: Tax rates minimising the heterogeneity of costs across Member States.

	Report	Uniform luxury VAT				Luxury taxed by category			
	rates	Sq. dev.	Minimax	Frugal	Max. gain	Sq. dev.	Minimax	Frugal	Max. gain
Tax rates (%)									
DST	3	12	7.7	12	12	12	6.3	12	12
Ad tax	15	50	60	0	60	36	60	18	60
FTT	.5	.59	.63	.49	1.7	.74	.56	.38	1.7
Wealth tax	.5	.51	.66	.99	0	0	.49	.27	0
Aviation tax	20	4.9	6.9	36	24	2.7	3.1	0	24
Luxury tax	20	3.7	0	0	0				
automotive	20					40	5.4	35	0
yachts	20					15	33	29	0
design & furniture	20					80	80	80	0
gourmet & dining	20					65	0	80	0
personal goods	20					0	0	5.9	0
hospitality	20					0	0	0	0
wines & spirits	20					0	0	0	0
RMSD from mean (% GNI)	.31	.152	.183	.219	.212	.12	.169	.141	.212
Max. cost (% GNI)	1.628	1.02	.884	1.266	1.323	.912	.852	1.039	1.323
Highest-cost country	CY	CY	LU	CY	NL	CZ	LU	LU	NL
Highest-cost frugal country	AT (.75)	NL (.85)	NL (.87)	AT (.7)	NL (1.32)	NL (.9)	NL (.85)	AT (.7)	NL (1.32)
New funds (EUR bn)	29.8	18.8	20.9	31.1	0	14	20.5	24.1	0

Note: Total new revenues are held at €103 bn (.705% of EU GNI), and each rate is capped at 4 times the report's rate. The criteria are: minimising the sum over the Member States of squared deviations of costs from the EU average (Sq. dev.), minimising the maximum cost (Minimax), minimising Sq. dev. under the constraint that the frugal countries (AT, DE, IE, NL, SE) pay less than the EU average (Frugal), and maximising the minimum State budget gain (Max. gain).